Fiscal Year:	FY 2011	**Task Last Updated:**	FY 07/23/2011
PI Name:	Trivedi, Rohit Ph.D.		
Project Title:	Dynamical Selection of Three-Dimensional Interfacial Patterns in Directional Solidification		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	06/01/2007	**End Date:**	11/30/2012
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date is now 11/30/2012 (Ed., 6/22/2012) NOTE: End date changed to 5/31/2012 (from 5/31/2010) per NSSC info dtd 9/2009 (10/2009) NOTE: Received NCE to 5/31/2010 (from 5/31/2009) per NSSC info dtd 6/2009 (10/2009)		
Key Personnel Changes/Previous PI:	Dr. Min Xu left the research group last year (during FY2011--Ed.).		
COI Name (Institution):			
Grant/Contract No.:	NNX07AK69G		
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Task Description:	The proposed research will consist of accurate and rigorous numerical modeling of the dynamics of pattern evolution, coupled with the results from low gravity experiments and critical ground-based studies in a model transparent system in which the dynamics of pattern evolution can be observed in situ. Initial results from the space station were analyzed to characterize the dynamical evolution of spatially extended patterns. New observations are described. A detailed quantitative phase-field model has been developed and applied to predict new observations from ISS, specifically the observation of oscillatory mode and the formation of multiplets. To obtain experimental data under diffusive growth conditions, ground-based experiments were carried out in thin samples of circular and rectangular cross-sections. The difference in pattern characteristic for a single cell/dendrite, single array of cell/dendrite and three-dimensional array is discussed.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The development of fundamental quantitative knowledge of microstructure evolution under dynamic growth conditions is crucially important for improving large scale industrial processes from casting to welding. The results will provide basis to predict and design reliable processing conditions that optimize microstructures and hence material properties and performance. These results are crucial in improving castings for aerospace and defense industries. In addition, the knowledge gained from this study will be applied to other pattern forming phenomena that occur in different fields of science such as combustion, fluid physics, cryo-biology, and food science. The benchmark data obtained under low gravity conditions will be a cornerstone for the validation of phase-field approach through unique quantitative comparison of model predictions and benchmark experiments in three dimensions.
Task Progress:	During this reporting year we started the experiments on the International Space Station. We shall thus describe some of these results from the Space Station first, and then show the development of a theoretical model to analyze these experimental observations. Finally, we shall describe some of the ground-based study on thin sample in which solidification occurs under diffusive growth conditions. I. Results from experiment on International Space Station Cellular and dendritic microstructures formed during solidification processes such as casting, welding, and brazing have a profound influence on the mechanical properties of a wide range of commercial metallic alloys of vital importance for the automotive, aerospace, and other industries. One of the critical parameters relevant to these technologies is the spatial arrangement of cells and dendrites in three-dimensions that influence the solute segregation patterns in the intercellular and interdendritic region. From a fundamental standpoint, dendritic microstructure development is controlled by a delicate interplay between anisotropic capillary forces acting on the nanometer scale of the solid-liquid interface and macroscopic transport of heat and solute on much larger scales, which is strongly influenced by convection under terrestrial conditions. Directional solidification experiments under microgravity conditions have provided a unique setting to investigate spatially extended three-dimensional array of cells and dendrites under diffusive growth conditions. Under terrestrial conditions, convection effects cause a significant variation in microstructure from the center to the edge of the sample, with cellular spacing decreasing towards the wall and a planar front stabilizes near the wall. Our experimental results under microgravity conditions, however, show a uniform microstructure along the cross-section that indicates that diffusive growth conditions are present in the bulk sample. Experiments under low gravity conditions have been conducted on the International Space Station using the Directional Solidification Insert in the DECLIC facility. Succinonitrile-0.24 wt% camphor alloy was used in this study since it is transparent so that the dynamics of three-dimensional interface pattern evolution under different growth conditions were visualized in situ, and the interface was visualized from images from two cameras that viewed the interface pattern from the side and from the top of the sample. In addition, the laser interference technique was used to reconstruct three-dimensional shapes of cells and dendrites near the tip region. All images are recorded on a disc. Experiments have been carried out over a range of velocities to obtain cellular and dendritic arrays. The time evolution of the interface pattern is characterized from the initial planar interface, to the onset of instabilities, dynamics of spacing adjustment and the final spatial arrangement. It is found that the final spatial arrangement is close to a hexagonal arrangement with some disorder. However, more complex arrangements in which both hexagonal and square array are present in the same experiment are also observed, and these are correlated with the presence of a small angle boundary in the initial solid seed. The time evolution of primary spacing is characterized by measuring the distances between the centers of the cells, and a distribution curve is obtained. The deviation from a perfect hexagonal arrangement is also determined by using the minimum spanning tree approach that gives the average distance and the deviation from the perfect hexagonal array. The variation the average primary spacing with velocity was characterized along with the distribution of spacing at each velocity. Note that he spacing decreases continuously with velocity when the transition from cells to dendrites occurs. This is in contrast to the spacing variation in a linear array grown in thin samples at 1g, where a sharp increase in spacing is observed, and theoretically predicted, at the cell-dendrite transition, as we will show in the section on ground-based studies. We have predicted this difference in spacing behavior between the linear array and a three-dimensional array by using the phase-field model. . Another important observation from low gravity experiments is the formation of multiplets in three-dimensions. Although multiplets have been observed in a linear array in thin samples, no observation in three-dimensional array has been seen before. In this case, some cells break-up into few cells with smaller spacing and smaller amplitudes. Although multiplets have been observed in thin samples, this is the first observation of multiplets, and the nature of these multiplets, in three dimensional growth. Finally, a novel oscillating mode during the cellular growth at low velocity is observed in low gravity experiments that shows that the oscillatory behavior is governed by the diffusive growth process. In a hexagonal cellular array, at V= 1.0 μm/s, three groups of cells, each group containing three cells, show coordinated oscillations in time with periodic change in the amplitude and cross- cross- sectional area. A phase difference of 120o is observed between these three groups. A detailed theoretical analysis of this oscillatory phenomenon has been carried out and presented in the next section. The results described above are obtained only from the first set experiments and from the limited number of images do. Significantly more data will be analyzed when all images downloaded in real time from ISS.

II. Theoretical modeling

We have carried out the first quantitative comparison of directional solidification experiments conducted on the International Space Station, where a transparent succinonitrile (SCN)-camphor organic alloy was used for in-situ visualization of the solid-liquid interface dynamics, and three-dimensional phase-field modeling of those experiments. The simulations exploit a quantitative phase-field model of the directional solidification of a dilute binary alloy and a massively parallel implementation of this model on graphics processing units (GPU). While previous phase-field simulations have been limited to two dimensions or small system sizes in three dimensions, the present GPU implementation has enabled us to simulate spatially extended systems of many cells in three dimensions on experimentally relevant length and time scales, with 1 hour of computation time corresponding approximately to 1 hour of experimental time. The phase-field simulations to date have allowed us to characterize a novel experimentally observed oscillatory mode of the cellular structure. The simulations have reproduced several salient features of this oscillatory growth mode. Firstly, the oscillations occur on a slow diffusive time scale of ~ 49 minutes that is in excellent agreement with observed period of ~ 45 minutes in the ISS experiments; microgravity played a crucial role in uncovering the existence of these oscillations that are normally altered or masked by convection effects under terrestrial conditions. Secondly, also as experimentally observed, the oscillations involve a coordinated motion of three groups of cells of the underlying hexagonal array structure that oscillate with a 1200 phase sift with respect to each other. The model has quantitatively predicted these observations. Ongoing work focuses on understanding the mechanism of those oscillations and how they are influenced by the structure of the array and growth parameters.

III. Ground-based experiments in thin samples

Two sets of experiments were carried out in ground-based study in thin samples where diffusive growth is obtained. In the first approach, directional solidification studies are carried out in thin circular capillary tube in which only a single cell or dendrite forms. In this case the primary spacing is fixed by the diameter of the tube, and the pattern evolution occurs under this constraint. This study allows one to explore steady-state solution for a given primary spacing, which cannot be examined in an array growth in which the primary spacing is selected by the dynamics of array growth. In the second approach, array dynamics are examined through the directional solidification in thin rectangular sample. In this case, the primary spacing selection plays a crucial role and the pattern characteristics are governed by the spacing selection. We shall discuss similar as well as significantly different observations in these two approaches and in the three-dimensional array observed in microgravity experiments. Directional solidification studies were carried out in circular capillary tubes of 100 µm and 200 µm ID, in which only one cell or one dendrite was obtained so that the primary spacing is constrained to correspond to the sample diameter. In the second set of experiment, directional solidification was carried out in thin rectangular samples in which an array of cells or dendrite is obtained. We first examine the results in thin circular capillary samples. In both samples, the shapes at very low lower velocity are cellular with a rounded tip shape, and with increase in velocity, they gradually become sharper and approach a parabolic shape and onset of sidewise instabilities occur that amplifies to form dendrites. The main difference in the two different size samples is that the side branches appear at a velocity of 13.0 µm/ in 100 µm ID sample, compared to about 3.0 µm /s in the 200 ID sample. The velocity at which sidewise instabilities initiate is roughly four times smaller in 200 ID sample compared to that in the 100 ID sample, which indicates that the velocity at which sidewise instability occurs is inversely proportional to the square of the sample diameter, or the primary spacing.

We now compare the results of a single cell or dendrite in a capillary sample with the results in an array growth obtained in a thin rectangular sample. In the thin rectangular sample, the primary spacing is not imposed so that there is a selection of spacing in addition to the selection of the morphology, tip radius and tip temperature. The tip radius is found to decrease as the velocity is increased, and it also decreases as the temperature gradient is increased at a given velocity. Two distinct branches of solutions are observed, with the low velocity branch corresponds to cellular shapes, while the higher velocity branch corresponds to the dendrite shape, and more importantly these two branches are not continuous and a discontinuity in the tip radius is observed at the transition velocity. The transition from the low velocity to high velocity branch occurs at a velocity at which the onset of sidewise instabilities occurs, which depends on the sample diameter and the thermal gradient. In contrast to single cell or dendrite, the array results show a continuous decrease ion tip radius with velocity and no discontinuity at the cell-dendrite transition is observed.

Bibliography Type:	Description: (Last Updated: 04/03/2024)
Articles in Peer-reviewed Journals	Bergeon N, Ramirez A, Chen L, Billia B, Gu J, Trivedi R. "Dynamics of interface pattern formation in 3D alloy solidification: first results from experiments in the DECLIC directional solidification insert on the International Space Station. (From the issue entitled 'Special Section: Abbaschian Festschrift')" Journal of Materials Science, 2011 Oct;46(19):6191–202. http://dx.doi.org/10.1007/s10853-011-5382-2 , Oct-2011
Papers from Meeting Proceedings	Pont G, Barde S, Bioulez Ph, Blonde D, Zapploi B, Garrabos Y, Lecoutre C, Beysens D, Bergeon N, Billia B, Noel NM, Ramirez AZ, Trivedi R. "DECLIC, First results on orbit." 61st International Astronautical Congress, Prague, Czech Republic, September 27-October 1, 2010. Proceedings, 61st International Astronautical Congress, Prague, Sept. 2010. Red Hook, NY : Curran Associates, Inc., June 2011. Paper IAC-10.A2.5.1. , Jun-2011